VANESSA LAU

San Diego, CA, 92092 | 323-704-9354 | valau@ucsd.edu | https://github.com/vmlau

EDUCATION

University of California, San Diego - *Data Science Major* San Diego, CA March 2027

- Current UC GPA: 3.958
- Minor: Cognitive Science

EXPERIENCE & EXTRACURRICULAR ACTIVITIES

Turning Green - *Data & Impact Research Intern* Jan 2026 - Present

- Led **mixed-methods longitudinal analysis** of multi-year Project Green Challenge datasets, integrating statistical and thematic approaches to identify cross-cohort trends and impact narratives
- Cleaned, transformed, and standardized **complex real-world datasets** to improve data integrity and analytical reliability
- Produced **executive-facing reports and visualizations**, distilling complex analyses into clear, actionable insights for **non-technical organizational leadership**, including the Executive Director, Program Administrator, and Advisory Board
- Collaborated cross-functionally to align analytical methodologies and **standardize reporting processes** across teams

UCSD Data Science Student Society | Natural Disasters Analysis Project -
Role : *Data Scientist* Achievement : 3rd Place Jan 2025 - Jun 2025

- Earned **3rd Place** in the 2025 DS3 Winter Projects Cohort (Advanced-level Climate Track) for analysis of storm trends.
- Processed, downsampled, and analyzed **>120 million atmospheric, geospatial, and hydrological datapoints** from ERA5, NOAA, LiDAR datasets to study storm trends (2005–2024)
- Engineered and tuned **Random Forest and Gradient Boosting models** (using Optuna) to classify storm types; evaluated performance via 5-fold cross-validation (ROC AUC, F1-score, Confusion Matrix, etc.)
- Applied **DBSCAN clustering and PCA** to uncover spatial and temporal patterns in storm behavior, delivering insights into climate change impacts

UCSD COGS 108 | Academic Rigor Analysis -
Role : Data Scientist Grade : A+ Apr 2025 - Jun 2025

- Analyzed **10+ years of student-reported academic data** to evaluate perceived rigor across UCSD majors using **regression and comparative statistical methods**
- Identified discrepancies between perceived and measured academic difficulty, challenging prevailing **STEM vs. Humanities assumptions**
- Communicated findings through **data visualizations and presentations** to foster discussion on academic bias and equity8

TECHNICAL SKILLS

- **Coding Languages:** Python (pandas, scikit-learn, xarray), SQL, Java, JavaScript, regex, HTML, CSS, R
- **Modeling & Analysis:** Regression, Classification, Random Forest, Gradient Boosting, DBSCAN, PCA, Statistical Testing
- **Data & Visualization:** Geospatial Analysis, Data Wrangling, Thematic Analysis, matplotlib, seaborn, D3
- **Tools:** Git, GitHub, VS Code, PostgreSQL
- **Software:** Google Workspace, Canva, Graphic Design (Procreate, Adobe Photoshop), Adobe Premiere Rush
- **Certifications:** Microsoft Office Specialist (Word, PowerPoint, Excel)
- **Languages:** English (fluent), Cantonese (basic conversational), Spanish (basic)